

built-in powerful audio codex engine.

That is why many people now immediately think of using their Wolfson/Raspberry Pi setup as a more flexible, lower-cost wireless media streaming system.

**Pi on Air.** Low-power RF module is an addition to its existing wireless protocols. A member from DesignSpark.com—an online community sponsored by RS Components—says, “The RF module is a USB device to provide bi-directional link to any device supporting serial communications, like another Raspberry Pi. Frequency, bandwidth, power output and data rates can be configured to allow multiple devices to communicate with each other and to other RF devices without any interference.

“The RF technology employed in Pi board enables significantly greater range (up to 300m depending on the terrain) and lower power consumption than other wireless protocols such as Wi-Fi or Bluetooth. The host device can send and receive up to 180 bytes of data per packet to other hosts within range, making it particularly useful for ‘sense and control’ applications.”

## What is the Raspberry Pi Foundation up to?

The foundation recently announced Compute module designed for engineers so they can harness the complete power of the Raspberry Pi for embedded applications. It is quite small and comes in a 3.5cm×6.5cm size. The module is compact enough to be mounted on a SO-DIMM socket, which is found in most motherboards these days. Once this module is plugged into a motherboard it would act as a stand-alone system and could be used as an add-on card for any particular application that customers might be developing.

A member from DesignSpark.com—an online community sponsored by RS Components—says, “Most of the times in final applications a designer may not need a complete Pi (complete Raspberry Pi board with all its connectors and interfaces), he may only need a slice of it. Compute module may be that slice, with

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a very small form factor of SO-DIMM.

“The good thing about the Compute module is that it retains the processing power of BCM2885 SOC with 512MB RAM and on-board Flash storage but outsources all its peripherals to companion I/O board. This makes it very easy for many Raspberry Pi users to switch to Compute module for their final applications.

“The Compute module lets designers explore full flexibility of BCM2885 SOC because of the availability of more GPIOs and interfaces. And the companion I/O board provides access to processor interface in much better manner, giving liberty to designers to make custom systems with ease. Besides, the compute module will encourage many designers to come up with a variety of companion I/O boards having different interfaces, peripherals and applications.”

Pallab Maji, senior research fellow, NIT Rourkela says, “While we were working on a robot project based on Pi, we found that certain gesture-controlled applications were required for which we had to connect an Arduino board to Raspberry Pi for a better performance. I think the Compute module is going to bring many different in-built capabilities for the designing aspect, like the increased number of interfacing options along with a better processing speed.”

## Is Raspberry Pi the only one capable of fueling innovation?

There is great competition between the different development boards available today, and this forces us to question ourselves: Which development board should be used and why?

Abhishek Rao, says, “One competitor, the RIoT board, mainly focuses on markets like mobile and tablet whereas BeagleBone is looked up to for more of industrial applications due to its ability to provide more number of I/O (input or output) functionalities required for such applications. If we look at

the Raspberry Pi, its compatibility with Arduino enables more than 300 add-on shields which can be used to give this board an edge over all the other boards in terms of cost, and in terms of applications that can be realised.”

A report by Mouser Electronics suggests that the Galileo, a product of Intel’s effort with Arduino, does not allow us deeper control of the processor. This report also suggests that ARM is the most popular processor with multiple tools and licensed by many, which is not the case with Intel’s products.

But Michael Leonard in one of his reports at Makezine suggests that despite built-in graphic support, BeagleBone Black is not powerful enough to support 1080p. However, if we look at the facts and features of Raspberry Pi, it has an integrated graphical Video core processor, composite video output and a full-size HDMI connector. Its capability of HDMI audio output and addition of Arduino compatible shields and components leaves no options for Michael Leonard but to support Raspberry Pi over others in a report by him.

Pallab says, “I have an old monitor, and when I connect that monitor to BeagleBone, certain things need to be taken care of. Whenever I connect the Beagle board to a VGA converter for a display on the monitor, I do not meet with much success. Since Beagle board does not provide the necessary +5V power to the device connected to the HDMI port, I have to provide an external power to the converter. Second, special cable needs to be considered as Beagle board has a micro HDMI port whereas my monitor has an HDMI port.” ●

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